AIOPS-02: Anomaly Detection

> **Series:** AIOPS — Dynatrace Intelligence | **Notebook:** 2 of 8 | **Created:** May 2026 | **Last Updated:** 07/30/2026

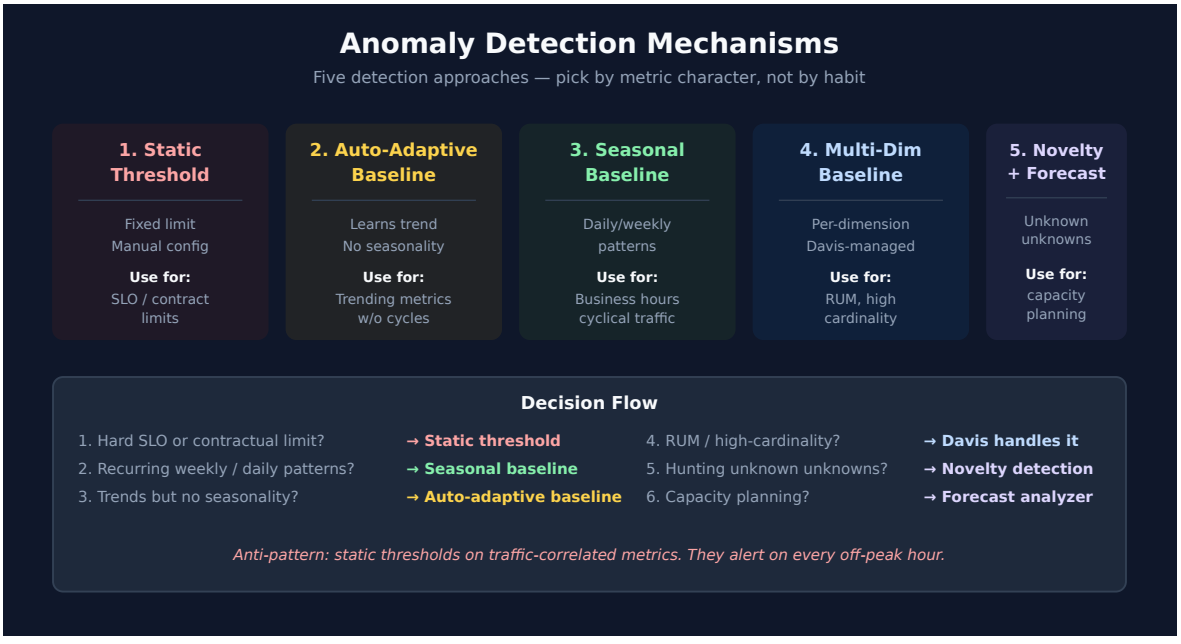
Overview

Anomaly detection in Dynatrace is **Predictive AI in action** — the platform learns what normal looks like and surfaces deviations. Five mechanisms cover the practical detection landscape: static thresholds, auto-adaptive thresholds, seasonal baselines, multi-dimensional baselines, and novelty / forecasting.

Most teams over-rely on static thresholds and miss what adaptive and seasonal detection would catch. This notebook walks the five mechanisms, when to use each, and how to test them against your data using the Davis analyzer MCP tools.

Audience: Platform admin tuning detection; SRE writing custom alerts.

Outcome: A working understanding of which detector to pick for each metric type, and live examples against your tenant.



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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS Gen3 with Anomaly Detection app installed
Permissions	`davis:analyzers:execute`, `settings:objects:read/write`, `events:read`
MCP	Dynatrace MCP server (for AIOPS-04 / AIOPS-06 integration); analyzers also exposed in the app
Optional	Monaco / Terraform for config-as-code (see AUTOM-05/06)

1. The Five Detection Mechanisms

1.1 Static threshold

Hard limit: *response time must be < 1 s*. Trips when the metric crosses the line. Configured manually. Fast to set up, but brittle — a threshold that fits Tuesday at 3 AM rarely fits Friday at noon.

Use when: an SLO, contract, or capacity ceiling defines the limit. Don't use for anything that varies with traffic.

1.2 Auto-adaptive threshold

Davis learns the metric's baseline and shifts the detection threshold as the baseline moves. No seasonal awareness — purely adaptive to recent trend.

Use when: the metric trends over time but doesn't have weekly / daily seasonality. Service throughput on a steadily-growing app is a classic fit.

1.3 Seasonal baseline

Davis learns the metric's daily, weekly, and (where it has data) yearly seasonality. Detection considers the *expected* pattern — Tuesday at 3 AM and Friday at noon get different thresholds.

Use when: the metric has obvious recurrence — business hours, weekday/weekend differences, monthly billing cycles.

1.4 Multi-dimensional (automated) baseline

Davis builds baselines per-dimension automatically — per region, per browser, per OS, per user-action. You don't configure this; the platform does it under the hood for RUM-like metrics.

Use when: you don't — Davis chooses. Your job is to let the high-cardinality dimensions through to it (don't pre-aggregate them away).

> **This is not a licence to split *your own* detectors by a high-cardinality dimension.** Davis's multi-dimensional baseline consumes the dimensions internally and still emits grouped problems. A custom detector grouped by a volatile dimension — application version, pod name, HTTP status code — emits a separate alert *per dimension value*: volume then scales with your deployment frequency, and alerts strand themselves on entities that no longer exist. Feed dimensions to Davis; do not fan your own alerting out across them. ALERT-02 §3 carries this as an anti-pattern.

1.5 Novelty detection and forecasting

Novelty flags patterns the model has never seen — sudden new error types, never-before metric shapes. **Forecasting** projects a series forward, useful for capacity planning and trend

extrapolation.

Use when: you're hunting for *unknown unknowns* (novelty) or projecting growth (forecast). Both available as Davis analyzers — see Section 5.

2. Picking the Right Detector

A simple decision flow:

1. *Is there a hard contractual or SLO limit?* → **Static threshold.**
2. *Does the metric have recurring weekly/daily patterns?* → **Seasonal baseline.**
3. *Does it trend over time without strong seasonality?* → **Auto-adaptive threshold.**
4. *Is it RUM / high-cardinality user-facing?* → **Davis handles it; don't pre-aggregate.**
5. *Are you hunting unknown unknowns?* → **Novelty detection.**
6. *Capacity planning?* → **Forecasting.**

Anti-pattern alert: static thresholds on traffic-correlated metrics. They alert on every off-peak hour and on every traffic spike. Move to seasonal or auto-adaptive.

3. Configuration Surface: App vs. Settings vs. Code

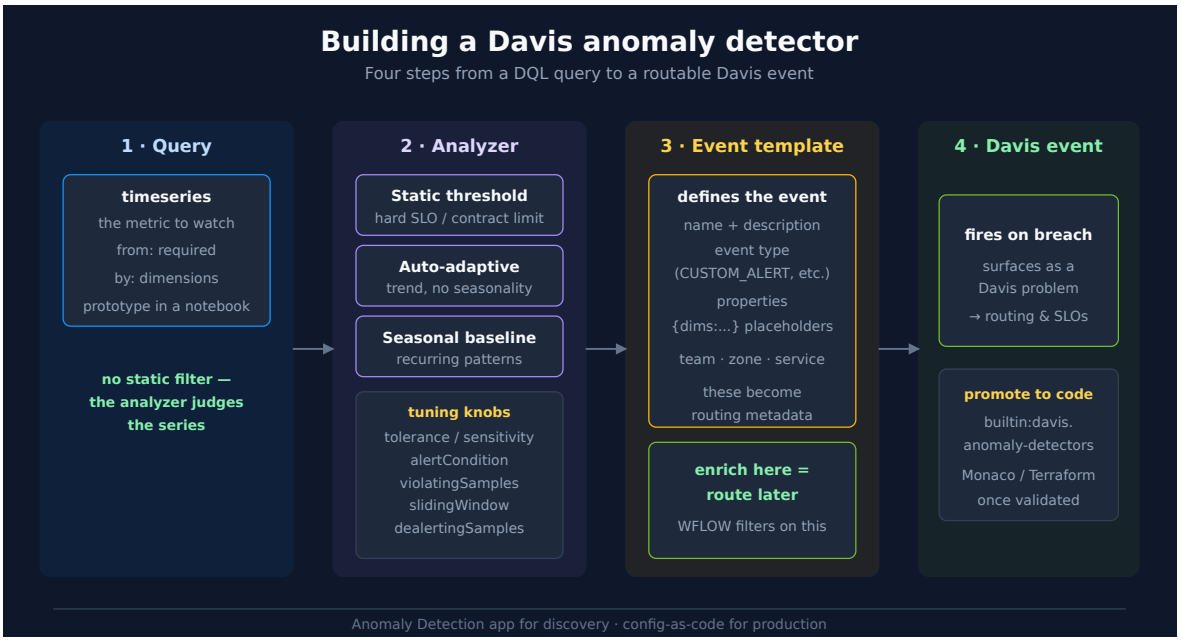
Three places to configure detection — pick one per environment, not one per detector.

Surface	When to use
Anomaly Detection app	Exploration, one-off detectors, quick wins. Best for SREs in the moment.
Settings 2.0 (UI)	Steady-state detection that's been validated; lives under specific settings schemas.
Config-as-code (Monaco / Terraform)	Anything you want versioned, reviewable, and reproducible across environments.

Production teams should converge on config-as-code. The app is the right place to *discover* what to alert on — but the moment a detector matters in production, it should live in source control. See **AUTOM-05** and **AUTOM-06** for the patterns.

4. Building a Davis Anomaly Detector

Sections 1–3 told you *which* mechanism to pick and *where* to configure it. This section walks the actual build in the **Anomaly Detection app** — the four steps that turn a DQL query into a routable Davis event.



Step 1 — The query contract

The detector samples a `timeseries` query on a schedule. Critically, **you do not write a `filter ... > threshold` in the detector query** — the analyzer (Step 2) is what decides whether the series is anomalous. Your query's job is to *return the metric*, grouped by the dimensions you want to alert per (`by:{...}`). A `from:` range is mandatory.

> **Develop the query in a notebook first.** This is the notebook-as-scratchpad pattern: write and run the `timeseries` here, confirm it returns the shape you expect, then paste it into the detector. A query that returns nothing in a notebook will silently never fire as a detector.

Step 1a — Records-based detectors for sparse signals

Not every signal is a metric series. For a condition that lives in raw logs or events and occurs rarely — failed logins, a specific exception, a licence error — set the detector's **Data type** to **Records** instead of a metric, and give it a deliberately large aggregation window:

```
fetch logs, from:now() - 60m
| filter matchesPhrase(content, "authentication failed")
| summarize failed_login_count = count()
| filter failed_login_count &gt; 500
```

With a 60-minute aggregation window, set the detector's **Delay** to **60 Minutes** so the evaluation window and the execution cadence line up. Two things improve at once:

- **Noise.** A sparse signal evaluated minute by minute produces false positives from ordinary clustering. Five failed logins in one minute is nothing; 500 in an hour is something. The window *is* the tuning.
- **Cost.** The detector runs once an hour rather than sixty times — a 60× reduction in evaluations against raw log data. Querying raw records on every evaluation is the expensive shape (FAQ-09, FINOPS-03), and widening the window is the cheapest mitigation short of extracting a metric at ingest (OPIPE).

> **Prototype this one especially carefully.** The query above is syntactically valid and executes, but on the tenant it was tested against `"authentication failed"` matched nothing, so `summarize` returned a single row with `failed_login_count = 0` and the final `filter` removed even that. Zero rows is precisely the silent-detector failure described above — the detector would install cleanly and never fire. Run the query *without* the threshold filter first and confirm the count is non-zero before you trust the phrase.

If the condition recurs often enough to warrant continuous watching, extract it to a metric in OpenPipeline and put an ordinary metric detector on that instead. Records-based detection is the right tool for genuinely sparse conditions, not a general substitute.

Step 2 — The analyzer and its tuning knobs

Apply one analyzer to the series:

Analyzer	Judges against	Use when
Static threshold	A fixed line	A hard SLO / contract / capacity ceiling
Auto-adaptive threshold	A learned, drifting baseline	The metric trends but has no seasonality
Seasonal baseline	A confidence band around the seasonal pattern	The metric recurs (business hours, weekly cycles)

The tuning parameters that control noise:

Parameter	What it does
`tolerance` (seasonal) / sensitivity	Width of the confidence band — higher = fewer alerts
`alertCondition`	Direction that trips: `ABOVE`, `BELOW`, or `OUTSIDE`
`violatingSamples`	How many points in the window must violate to fire (max 60)
`slidingWindow`	How many points are evaluated together (max 60; ≥ `violatingSamples`)
`dealertingSamples`	Consecutive clean points required to close the alert (max 60)
`alertOnMissingData`	Treat absent data as a violation
query offset	Shift the evaluation window to absorb data latency

The `violatingSamples` / `slidingWindow` pair is your primary noise control — requiring, say, 3 violations out of a 5-point window suppresses single-spike flapping while still catching sustained breaches.

Step 2a — Problem-event trigger delay (a different layer)

Forthcoming / rolling out (SaaS 1.344). SaaS 1.344 released 07/27/2026 with a **staged tenant rollout** (from 07/29/2026): **problem-event trigger delays are configurable** — how long a Davis event must persist before it opens a *problem*. Verify it has reached your tenant before you plan around it.

This is not an analyzer parameter, and it is deliberately not a row in the table above. The two sit at different layers, and conflating them is how tuning goes wrong:

Layer	Decides	Configured in
Analyzer parameters (`violatingSamples`, `slidingWindow`, `dealertingSamples`, tolerance)	Whether *this series* is anomalous — i.e. whether a Davis event fires at all	Per detector, in the Anomaly Detection app or `builtin:davis.anomaly-detectors`
Problem-event trigger delay	How long a Davis event must persist before it is promoted to a *problem*	Platform-side, applying across events

So the trigger delay **complements** `violatingSamples` / `slidingWindow` rather than replacing it. Its distinct value is reach: because it applies platform-side, it damps flapping from detectors **you do not own** — built-in detections, and detectors configured by other teams — which no amount of analyzer tuning on your own detectors can touch.

For any detector you build yourself, `violatingSamples` / `slidingWindow` remains the control to rely on and the right first lever. Tune the detector to stop firing spurious events; reach for the trigger delay for noise arriving from events you cannot tune at source.

Step 2b — Scoping the detector with Segments

Alongside the analyzer, the Anomaly Detection app offers **Set scope** (Simple or Advanced tab) with a **Segments** field — choose one or more segments to restrict which slice of data the detector evaluates. This lets you alert on a specific subset such as a region, cluster, or environment without rewriting the query.

> **Do not duplicate a filter the segment already applies.** Segment filters are injected into the query at execution time, so re-stating them in the `timeseries` query is redundant and makes the detector harder to reason about.

This is the only place segments touch alerting. Segments scope *detection* — which signals get evaluated — and they scope *queries*. They do **not** scope workflow triggers, notification routing, or problem visibility. If you are migrating off Management Zones and expecting segments to carry your alert routing, see MZ2POL-09: routing moves to problem-triggered workflows filtered on affected-entity tags, not to segments.

Step 3 — The event template (this is what makes routing possible)

When the detector fires it emits a Davis event whose shape *you* define: an event **name** and **description** (both support `{dims:...}` placeholders that interpolate the breaching dimensions), an event **type** (`CUSTOM_ALERT` , `ERROR_EVENT` , `PERFORMANCE_EVENT` , …), and **properties** — key/value pairs such as team, zone, or service.

Those properties are the metadata a downstream workflow filters on. **Enrich here or you cannot route later** — a detector that fires a bare event with no team/zone property forces every workflow to re-derive ownership from the affected entity. Spend the effort in the template.

Attribute the event to a real entity, or it can never correlate. Alongside name, description, type and properties, a Davis event carries `dt.smartscape_source.id` — the Smartscape entity ID of whatever the signal is *about*. Davis's universal correlation rule merges every event

naming the same entity into a single problem (AIOPS-03 §1). Set it to an actual host, service, or workload ID, normally interpolated from a `by:{}` dimension the query already groups on.

A detector that leaves this unset — or fills it with a free-text label — matches nothing, so it merges with nothing: **every firing opens its own problem.** No amount of threshold tuning fixes that, because the noise is structural rather than sensitivity-related. Section 8 has a query that finds these in your own tenant, and a real example of one firing thousands of times a week.

Two event types that never open a problem. `CUSTOM_INFO` and `WARNING` are both severity SEV-5: they are stored in Grail, are fully queryable, and can trigger workflows, but they do not raise problems. That makes them useful for chronic issues already tracked elsewhere, for routing an observation to Slack or Jira without cluttering the Problems app, and — most valuably — for calibration:

> **Shadow-deploy a new detector.** Rather than guessing a threshold, ship the detector with `event.type` set to `CUSTOM_INFO`, leave it running for two to four weeks, then measure how often it *would* have fired (Section 8). Tune against that evidence, and only then promote it to `CUSTOM_ALERT`. This turns threshold-setting from a guess into a measurement, and being wrong during the observation window costs nothing but storage.

Step 4 — Fire, then promote

On breach the event surfaces as a Davis problem and flows into routing (WFLOW series) and SLO error budgets. Once a detector matters in production, move it out of the app and into config-as-code (`builtin:davis.anomaly-detectors` — see Section 6) so it is versioned and reviewable.

> **Sources:** Anomaly detection configuration (DT docs), Set up anomaly detectors via API (DT docs).

5. Custom Alerts via DQL

Beyond pre-canned detectors, Anomaly Detection app supports **DQL-based custom alerts**. You write the query, the app evaluates it on a schedule, and a breach generates a Davis event.

Example — alert when a service's error rate breaches 5% for 5 minutes:

```
// Custom alert: service error rate &gt; 5% in the last hour
// (When wired into the Anomaly Detection app, this evaluates on a schedule.)
timeseries {
    failures = sum(dt.service.request.failure_count),
    total    = sum(dt.service.request.count)
  },
  by:{dt.smartscape.service},
  from:-1h, interval:1m
| fieldsAdd error_rate = (failures[] / total[]) * 100
| fieldsAdd avg_error_rate = arrayAvg(error_rate)
| filter avg_error_rate &gt; 5
| sort avg_error_rate desc
| limit 20
```

Watchpoints when writing custom-alert DQL:

- Always include a `from:` time range. Detectors that omit it are rejected.
- Use named-parameter `decimals:`, `then:`, `else:` where required (`round`, `if`, `substring`).
- Aggregations used downstream need explicit aliases (`error_rate = ...`, `mttr = ...`).
- Element-wise array operations on timeseries: `failures[] / total[]` returns an array; wrap in `arrayAvg()` (or similar) to collapse to a scalar before `filter`.

6. Metric Events and Settings 2.0 Schemas

Not every alert needs the DQL-based detector. **Metric events** are the Settings 2.0 mechanism for threshold / baseline alerting directly on a metric key — lighter weight than a DQL detector, and the right tool when you're alerting on a single pre-existing metric.

Mechanism	Settings 2.0 schema	Reach for it when
DQL-based Davis anomaly detector	`builtin:davis.anomaly-detectors`	The signal needs a query — joins, derived fields, multi-metric logic
Metric event	`builtin:anomaly-detection.metric-events`	You're alerting on one existing metric key with a threshold or baseline

Both are first-class config-as-code targets. Provision them with Monaco or Terraform exactly as in **AUTOM-05 / AUTOM-06**; the schema name is the `--settings-schema` / resource selector. This is how a validated detector from the app becomes a reviewed, version-controlled artifact.

> **Sources:** Metric events (DT docs), Anomaly detection configuration (DT docs).

7. Testing Detectors with Davis Analyzers (MCP)

The Dynatrace MCP server exposes Davis analyzers as callable tools. Useful when you want to test a detector against historical data before wiring it to a setting or a workflow.

MCP tool	What it does	Permission
`mcp__dynatrace__static-threshold-analyzer`	Test a static threshold against a series	`davis:analyzers:execute`
`mcp__dynatrace__seasonal-baseline-anomaly-detector`	Detect anomalies vs. seasonal pattern	`davis:analyzers:execute`
`mcp__dynatrace__adaptive-anomaly-detector`	Detect anomalies vs. learned baseline	`davis:analyzers:execute`
`mcp__dynatrace__timeseries-novelty-detection`	Flag never-seen patterns	`davis:analyzers:execute`
`mcp__dynatrace__timeseries-forecast`	Forecast future series; capacity planning	`davis:analyzers:execute`

Workflow pattern: write a `timeseries` query that returns the metric you care about, then pass the query to the analyzer. The analyzer's output is structured (anomalies, forecast bands, novelty flags) and can be embedded in workflows or notebooks.

All five analyzers also have GUI surfaces in the Anomaly Detection app — the MCP tools are the same engine accessed programmatically.

8. Anomaly Volume in Your Tenant

Davis events are the raw signals before Causal AI groups them into problems. Look at the category breakdown to understand what kinds of anomalies your detectors are firing.

```
// Davis event volume by category — last 24h
fetch dt.davis.events, from:-24h
| summarize count = count(), by:{event.category}
| sort count desc
```

```
// Active custom-alert problems (DQL-based custom alerts surface here)
fetch dt.davis.problems, from:-7d
| filter event.category == "CUSTOM_ALERT"
| summarize count = count(), by:{event.status}
| sort count desc
```

Reading the result: A high `CUSTOM_ALERT` count means your custom-DQL detectors are firing — review whether they're noisy. A high `RESOURCE_CONTENTION` or `ERROR` count means out-of-the-box auto-adaptive / seasonal detection is doing real work.

Finding your noisiest detectors

The category breakdown tells you what *kinds* of anomaly fire. This tells you which specific detectors, and — the part that matters most — whether each one is attributable and navigable:

```
// Noisiest non-informational detectors over 7 days.
// The two ID columns answer different questions - see the notes below.
fetch dt.davis.events, from:-7d
| filter not in(event.category, {"INFO", "WARNING"})
| summarize alerts = count(), by:{event.name, dt.settings.object_id,
dt.smartscape_source.id}
| sort alerts desc
| limit 20
```

Real output from a demonstration tenant over seven days, abbreviated to the top rows:

`event.name`	`alerts`	`dt.settings.object_id`	`dt.smartscape_source.id`
`YOU ARE A PROBLEM`	5,378	*null*	*null*
`Response time degradation`	2,439	*null*	`SERVICE-6C36694E683AD694`
`Failure rate increase`	1,022	*null*	*null*

`event.name`	`alerts`	`dt.settings.object_id`	`dt.smartscape_source.id`
`DYNATRACE USER LOGIN`	285	`builtin:davis.anomaly-detectors` …	*null*
`Backoff event`	254	`builtin:anomaly-detection.kubernetes.workload` …	`K8S_DEPLOYMENT-79081F0B98463CC2`

Read the top row first. A custom detector firing 5,378 times in seven days with **no `dt.smartscape_source.id`** is the anti-pattern from Section 4 in its natural habitat: unattributed, therefore unable to merge, therefore one problem per firing — roughly 32 problems an hour from a single misconfiguration. Re-tuning its threshold would not help. The event template has to name a real entity.

Then read the two ID columns, which answer different questions:

Column	Question it answers	Null means
`dt.smartscape_source.id`	Can this alert correlate with anything?	It cannot — each firing stands alone
`dt.settings.object_id`	Can I navigate to the configuration behind it?	No settings object; identify it by `event.name` instead

`dt.settings.object_id` is the Settings API `objectId`, so a populated value leads straight to the detector's configuration. A useful property: it encodes both the settings **schema** and the **scope**. The values abbreviated above resolve to `builtin:davis.anomaly-detectors` scoped to the tenant, and `builtin:anomaly-detection.kubernetes.workload` scoped to one specific `KUBERNETES_CLUSTER`. That scoping is why a single event name legitimately appears under several distinct object IDs — one per cluster — rather than indicating duplicated configuration.

> **Two caveats.** `dt.settings.object_id` is marked **experimental** in the semantic dictionary (`dt.smartscape_source.id` is `stable`) — fine for triage, not something to hard-code an integration against. And a seven-day unfiltered scan of `dt.davis.events` is not cheap: the run above scanned 12.9 GB, and the 30-day version below scanned 52.6 GB. Narrow the window for routine checks (FAQ-09, FINOPS-03).

For what counts as "too noisy" in the first place, ALERT-99 §3 carries the 0.1%-of-observed-time yardstick and the audit cadence that uses it.

Measuring what a shadow-deployed detector would have done

The calibration pattern in Section 4 needs evidence before promotion. This measures the firing frequency of a detector running in observation mode as `CUSTOM_INFO` or `WARNING`:

```
// How often would this observation-mode detector have fired?
// Replace the event.name with your own detector's name before running.
fetch dt.davis.events, from:-30d
| filter in(event.category, {"INFO", "WARNING"})
| filter event.name == "NR ALERT: Host CPU &gt; 80% sustained 5m"
| summarize firing_count = count(), by:{day = bin(timestamp, 24h)}
| sort day desc
| limit 10
```

Run against a ported static threshold held in observation mode, the ten most recent daily counts came back **236, 347, 459, 504, 470, 474, 502, 490, 480, 470** — the first of those a still-running partial day.

That detector must not be promoted. Roughly 470 firings a day would mean up to roughly 470 problems a day from a single rule — correlation would merge some of them, but nothing like enough — against a healthy-detector expectation of one or two firings a *month* (ALERT-99 §3). The evidence says retune before promoting: widen the sliding window, raise the threshold, or — since this is CPU on hosts, a traffic-correlated signal — move it off a static threshold onto auto-adaptive entirely (Section 1).

That is the whole value of the shadow deploy. Promoted straight to `CUSTOM_ALERT`, the same rule would have paged someone several hundred times a day before anyone discovered the threshold was wrong; held at `CUSTOM_INFO`, it cost nothing to learn the same thing.

> **Sources:** Avoid overalerting (DT docs), Anomaly detection configuration (DT docs). **Derived:** the promote/retune verdict applies the 0.1% yardstick to the measured firing counts; the `dt.settings.object_id` schema-and-scope property is an observation from the returned values, not a documented contract.

9. Cross-Series Pointers

- **AUTOM-05, AUTOM-06** — anomaly detection settings as code (Monaco, Terraform)
- **WFLOW** — once a detector fires, route the resulting Davis event into a notification or remediation workflow

- **DBMON, CLOUD, K8S** — domain-specific anomaly patterns sit in those series; this notebook is the canonical reference for the *mechanisms*
- **AIOPS-05** — the AI models behind these detectors (causal correlation, predictive baseline, seasonal)
- **AIOPS-06** — agentic patterns: scheduled analyzer runs in workflows

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